La Scuola d'Italia / Dr. Huson / IB Math 11 (a)
28 October 2026

Nickname: SOLUTIONS

## 2.6 Early Finishers: Lines, Exponents, Radicals and Sequences

*For students who finish the quiz early. A calculator is allowed.*

1. [Maximum mark: 11]

   (a) Solve the equation $25^{x+1} = 125^{2x-3}$. [3]

   (b) Find the integers $a$ and $b$ such that $\left(2+\sqrt{3}\right)\left(a+b\sqrt{3}\right) = 12+7\sqrt{3}$. [3]

   Consider the equation $3\left(9^x\right) - 82\left(3^x\right) + 27 = 0$.

   (c) By letting $u = 3^x$, solve the equation. [5]

(a) $5^{2(x+1)} = 5^{3(2x-3)}$

$2(x+1) = 3(2x-3)$

$11 = 4x$

$x = \frac{11}{4}$

(b) $2a + 2b\sqrt{3} + a\sqrt{3} + b(\sqrt{3})^2 = 12 + 7\sqrt{3}$

$(2a+3b) + (a+2b)\sqrt{3} = 12 + 7\sqrt{3}$

$2a+3b=12$ (1)

$a+2b=7$ (2)

$0a+1b=2$ $\quad 2\times(2)-(1)$

$a=3,\ b=2$

(c) $3(9^x) - 82(3^x) + 27 = 0 \quad u = 3^x$

$3u^2 - 82u + 27 = 0$

$(3u-1)(u-27) = 0$

$u = \frac{1}{3},\ u = 27$

$x = -1,\ 3$

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2. [Maximum mark: 8]

An arithmetic sequence has first term $u_1 = 7$ and common difference 3. A geometric sequence has first term $v_1 = 2$ and common ratio 1.5.

(a) Write down an expression for $u_n$ and an expression for $v_n$. [2]

(b) Find the least value of $n$ for which $v_n > u_n$. [3]

(c) Find the least value of $n$ for which the sum of the first $n$ terms of the geometric sequence is greater than the sum of the first $n$ terms of the arithmetic sequence. [3]

(a) $u_n = 7 + 3(n-1)$

$v_n = 2 \cdot (1.5)^{n-1}$

(b) $v_n = 2 \cdot (1.5)^{n-1} > 7 + 3(n-1) = u_n$

| $n$ | $u_n$ | $v_n$ |
|---|---|---|
| 7 | 25 | 22.8 |
| 8 | 28 | 34.2 |

$n = 8$

(c) $S_{u,n} = \frac{n}{2}(2 \cdot 7 + 3(n-1))$

$S_{v,n} = \frac{2(1.5^n - 1)}{1.5 - 1} = 4 \cdot 1.5^n - 4$

| $n$ | $S_{u,n}$ | $S_{v,n}$ |
|---|---|---|
| 9 | 171 | 149.8 |
| 10 | 205 | 226.7 |

$n = 10$